Line Monitoring of Selected Wolf-Rayet Southern Hemisphere Star Monitoring Campaign

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Abstract

We conducted a campaign of low resolution (R 1800) spectrographic observations of a selected list of brighter southern hemisphere Wolf-Rayet and O(f) stars using the FlexSpec1/600 and FlexSpec1/900 spectrographs with the Watson 24-inch CDK telescope located at the Obstech observatory complex in Chile. This paper reports some preliminary results from the on-going campaign. In this paper we discuss the observational procedure, some of the issues encountered, reduction approaches and techniques. The authors are grateful for the generous allotment of time with Dr. Stan Watson's telescope facilities.

1 Overview

A campaign of observing bright Wolf-Rayet stars, magnitudes < 11.0 and declinations < -5 degrees was conducted using the custom FlexSpec1 R~1100 spectrograph mounted to a Planewave CDK-24 on an L600 mount located at Obstech Observatory, Chile. Early in the project we added a R~150 spectrgraph in anticipation of observing fainter targets. Both instruments use a QHY 268M camera.

We share the telescope with top amateur LGRB imagers, and made our observations during the bright-time, 10-day interval around full moon. We start with targets in the west, work backwards towards the meridian, and spend the most time on the meridian. We observed our target stars on every clear night - airmass permitting.

As part of this campaign, we used available A0V stars as references. The reference stars are covered in the accompanying paper.

Targets were selected using a SIMBAD query for WR stars, conditioned and saved in a comma-separated-file. This file was loaded into TOPCAT, extended with the SIMBAD cross-match tool. Complete information for targets was written to a PostgreSQL database. A similar technique compiled the AOV stars. A subset of candidates based on magnitude, field contamination, accessibility to Obstech, Chile was created for use at the telescope. All coordinates are in decimal degrees, but carrying the sexigesimal format allowed interface to the Planewave 'Favorites' list. Recently we learned the python interface to Planewave scopes uses decimal degrees.

2 Observations

During observations, the two tables were sent to Aladin using the TOPCAT interop functionality sending sky coordinates and row indices. The ra/dec were cross tied to the tables such that selecting a star in the table would position Aladin to the field and vice-versa. In Aladin, target stars were denoted with large, solid red circles where reference stars were indicated with large, solid blue circles. The reference stars were chosen by zooming out in Aladin where the DSS color image was the default image used. This provided a subject guess to extinction for the WR stars and helped to avoid the inevitable crowded fields. Additional overlays of dim Gaia stars help reject candidates with possible slit contamination. The star pairs were recorded in one table together with the exposure time sequences.

The main computer for the telescope is a Windows 11 based environment to meet ASCOM ASCOM Initiative (2026) requirements for the equipment. Remote operations were done in real-time using the RVNC viewer. We later switched to TightVNC and Tailscale A JetKVM switch was added to the Chile machine allowing for a web-based interface to the main machine at the bios level. This permits un-wedging the machine during update scenarios that require human intervention before the VNC servers start.

Two shared folders were created allowing access of files accumulating in Chile by observers in Chile, Colorado and Arizona in the USA, and Newcastle Upon Type in England. This brings Linux based analysis tools to bear for quick checks of the data. We used SAOImage ds9 Joye and Mandel (2003) to examine images via the VNC link. This saved us early on, as the magnitudes of some emission features saturated, where the continuum was well within limits. In these cases we took short exposures to fix the flux of bright features and longer exposures to bring out lesser line details.

Recently we changed the R1100 spectrograph, to use a 1800 lmm grating giving a R3200.

3 Results

Preliminary reductions were performed with the Demetra Cochard (2026) software package, images below are from PlotSpectra Lester (2026). The packaged handled image acquisition and initial trimming using their crop image feature. Images are not binned. Early in the campaign we noticed we were saturating the large emission features, an event that was not well reported in Demetra. This was determined with the 'Pixel Analysis' table in SAOImage/DS9. For some targets short exposures were used to fully capture the emission features at the expense of smaller significant lines, followed by longer exposures to gain sufficient SNR in the smaller features.

Later processing followed the usual zero (bias) subtraction, extraction of the apertures one at a time and individually wavelength calibrated. The features were subjected to the IRAF Greenfield and White (2006) clean step, and we found boxcar windowing by 3 or three steps helped to clean the images, at the expense of resolution. We are conducting a more precise examination of images now, using continuum subtraction and flux calibration.

We found PHD2 Guider OpenPHDGuiding Project (2026) to be offer high repeatability in placing the target stars on the slit using their 'set and restore' function.

Observations reveal a wide, non-periodic and quasi-periodic variation in spectral features. Quasi-periodic results are from reported doubles where the period of variation covers periods of approximately one year.

4 Conclusions

These are preliminary results, presented here to promote the idea that WR and similar emission targets could be handled by use of narrow passband (10nm) filters for continuous monitoring at a much faster cadence, more targets per night, smaller apertures etc. When important changes are detected, we then bring the larger apertures to bear.

5 Acknowledgments

We appreciate the support from Dr. Stanley Watson for the 10 days of bright time observations over these past two years. Special thanks to Anthony Rodda for travel, installation and upgrades to the telescope. We extend special thanks to Ken Williams for his help with the remote communications and general IT support of the computer environment.

main_id	sp_type	ra_d	dec_d	u	b	v	r	i	j	h	k
HD 62910	WN6/WC4	116.24259	-31.90821	10.080	10.440	10.100	10.170		8.570	8.322	7.928
HD 63099	WC4+O7	116.46000	-34.33014	11.470	11.430	10.500	10.370		8.452	8.110	7.545
HD 76536	WC	133.74653	-47.59241	9.040	9.130	8.800	9.520		7.486	7.245	6.612
HD 115473	WC5	199.61664	-58.13711	9.090	9.680	9.000	9.720		8.411	8.206	7.555

Table 1: Characteristics of the sample results.

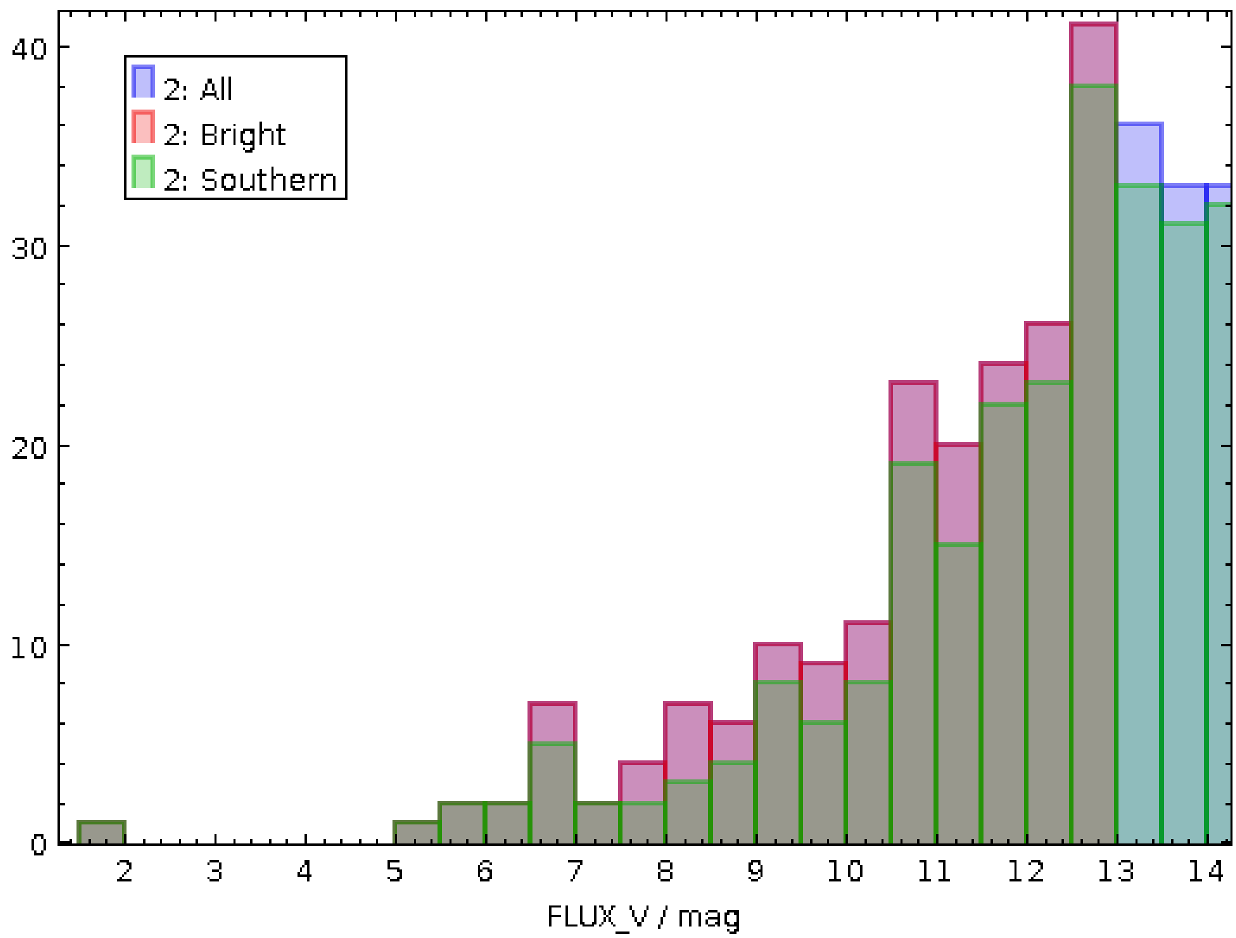

Figure -1: Histogram of WR stars in catalog.

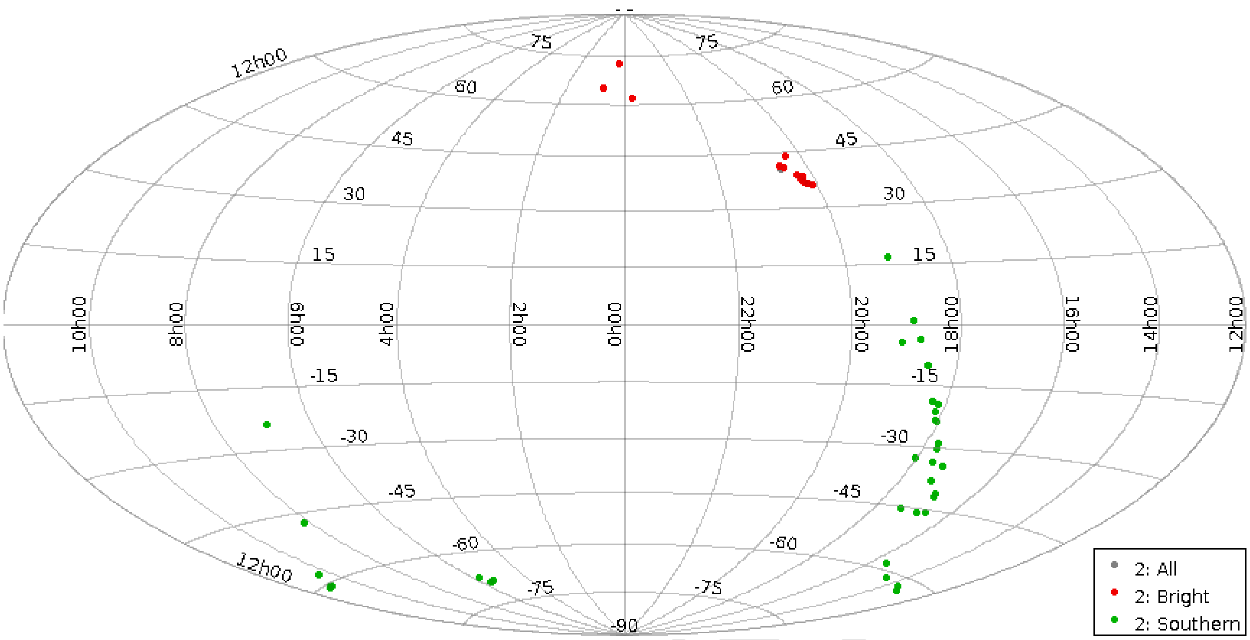

Figure -2: Sky location of 1,695 star.

A Preliminary Results

Observations range from 15 August, 2024 through 18 Feb, 2025 in these preliminary results.

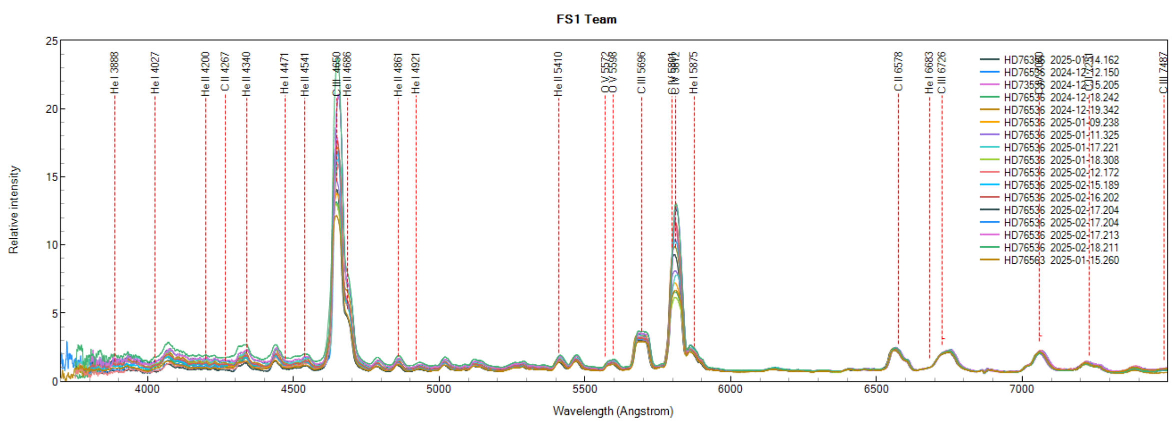

Figure A-3: HD76536 FS1 Team

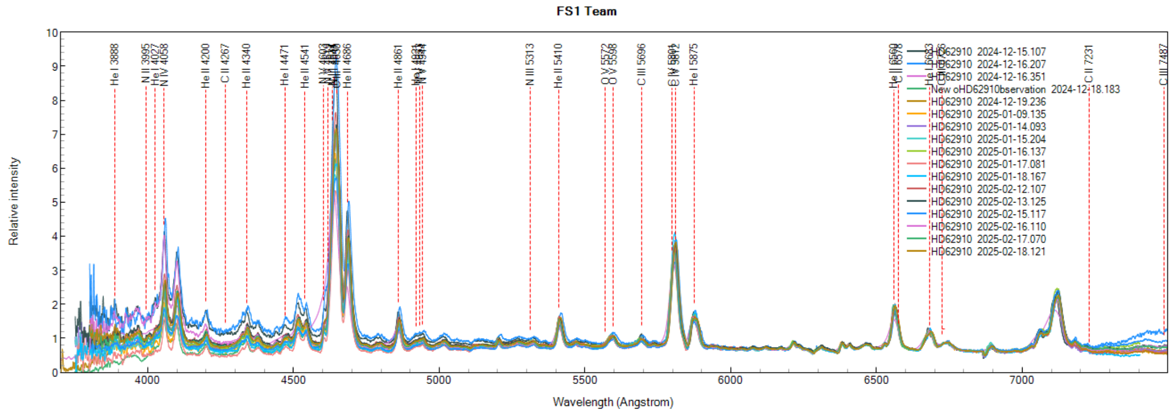

Figure A-4: HD62910 FS1 Team

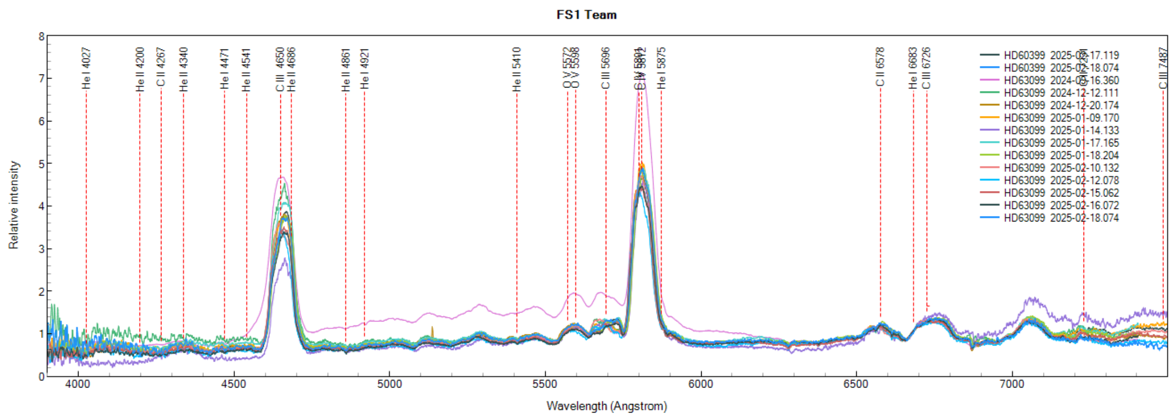

Figure A-5: HD60399 FS1 Team

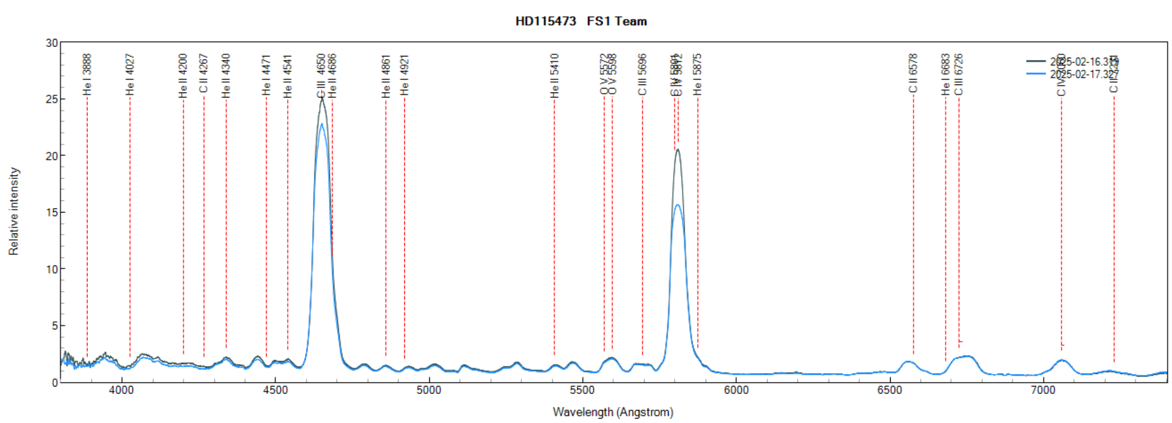

Figure A-6: HD115473 FS1 Team

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